

BREAK INTO AEROSPACE

From Control Theory to Aerospace

Career Transition Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Control theory specialists are among the most directly valuable engineers that aerospace can recruit — and among those who most consistently fail to communicate that value in the interview room. The failure is not technical. The Laplace transform, the root locus, the Bode plot, the LQR synthesis, the Kalman filter, the model predictive controller — these are the same mathematical objects whether they govern a chemical plant, a robotic arm, a power converter, or a flight control system. The failure is contextual: the control engineer who arrives at an aerospace interview able to design a state estimator but unable to explain what a phugoid mode is, what the short-period requirement means for handling quality, or why the structural mode notch filter creates a stability margin problem has the mathematical tools and not the physical domain in which to apply them. This guide is about closing that gap — specifically, precisely, and with enough technical depth that the aerospace interviewer recognises the preparation immediately.

The distinction between control theory as a mathematical discipline and control systems engineering in aerospace is substantial, and it matters. Academic control theory is largely domain-agnostic: a second-order transfer function with a natural frequency and damping ratio looks the same whether it represents a motor shaft, a pressure vessel, or an aircraft in pitch. Aerospace control engineering is deeply domain-specific: the short-period natural frequency of a commercial transport aircraft must meet MIL-HDBK-1797B Level 1 handling quality requirements; the phugoid mode must be stable for hands-off flight to be possible; the structural mode at the first wing bending frequency must be notched in the flight control law to prevent flutter excitation; and the control law must remain stable throughout the flight envelope as the aerodynamic derivatives change with altitude, Mach number, and configuration. The control engineer who understands what these requirements mean physically — not just how to tune a controller to meet a frequency domain specification, but why the specification has the physical value it does — is the engineer who contributes to flight control system design from the first day rather than after an extended domain learning period.

This guide is written for control theory specialists at all career stages — recent graduates with strong control theory foundations from electrical, mechanical, aerospace, or systems engineering programmes; mid-career control engineers from industrial automation, robotics, power systems, or process control backgrounds; and senior control systems engineers who have led control system development for non-aerospace applications and want to bring that leadership to the aerospace domain. It covers the full range of aerospace control applications — from flight control systems for fixed-wing and rotary-wing aircraft to spacecraft attitude determination and control to autonomous flight systems to engine control to guidance and navigation — because the control theory specialist's preparation differs by target application domain more than by career level.

The aerospace roles most directly accessible from a control theory background: flight control system engineer (the design, analysis, and verification of the control laws that govern aircraft response to pilot inputs and disturbances), spacecraft attitude determination and control systems engineer (ADCS — the control of spacecraft orientation using reaction wheels, thrusters, or magnetic torquers), guidance, navigation, and control engineer (GNC — the combined design of trajectory guidance, state estimation from sensors, and control law execution), autonomous flight systems engineer (the control architecture for unmanned aircraft, eVTOL autopilots, and urban air mobility autonomy), engine control systems engineer (the FADEC control laws that govern thrust, fuel flow, and engine protection systems), flight dynamics and stability analysis engineer (the open-loop analysis of aircraft dynamics that precedes control law design), aerospace-servoelastic analysis engineer (the coupled analysis of aerodynamics, structural dynamics, and flight control that determines flutter and structural mode interaction), and launch vehicle guidance and control engineer (the trajectory control and attitude stabilisation of launch vehicles during ascent).

The employer landscape: aircraft OEM flight control groups (Airbus Flight Control Systems in Toulouse, Boeing flight control in Seattle, Embraer), flight control system suppliers (BAE Systems flight control, Collins Aerospace flight control, Curtiss-Wright, Moog), spacecraft ADCS specialists (Surrey Satellite Technology, GMV, Airbus DS attitude control, Ball Aerospace, L3Harris), autonomous systems and UAV companies (Shield AI, Joby Aviation autonomy, Archer, Wisk, Zipline, Wing), engine control specialists (Rolls-Royce engine control, GE Aviation FADEC, Pratt & Whitney FADEC, Safran Electronics), guidance and navigation specialists (Honeywell navigation and guidance, Collins Aerospace navigation, Leonardo Finmeccanica navigation), aeroservoelastic analysis teams at OEMs (Airbus loads and aeroelastics, Boeing loads), and defence aerospace missile and weapon guidance groups (MBDA, Raytheon, L3 Technologies, Saab).

PART I: WHAT AEROSPACE EMPLOYERS ACTUALLY SEE IN A CONTROL THEORY SPECIALIST

The Genuine Advantages — What Transfers Directly

The control theory specialist's advantage in aerospace is more direct and more complete than any other discipline transition in this series, because the mathematical core of aerospace control — stability analysis, loop shaping, state estimation, optimal control — is identical across domains. The engineer who can derive the state space representation of a dynamic system, compute its eigenvalues to assess stability, design an LQR controller that minimises a quadratic cost functional, and implement a Kalman filter to estimate unmeasured states from noisy measurements has the complete analytical toolkit for flight control system design, spacecraft attitude control, and guidance and navigation. What they do not yet have is the aerospace physics that the mathematics must be applied to.

Linear systems theory transfers completely. Every classical aerospace control text — Bryson's *Control of Spacecraft and Aircraft*, McRuer's *Aircraft Dynamics and Automatic Control*, Stevens and Lewis's *Aircraft Simulation and Control* — uses the same state space formulation, the same transfer function analysis, the same frequency domain stability criteria as any control textbook. The Bode gain and phase margin requirements that aerospace uses to assess stability robustness are the same Bode margins used in any control system design. The root locus that shows how poles migrate as gain changes is the same root locus in an aircraft pitch control law as in a process control loop. The LQG/LTR methodology that achieves loop shaping while accounting for model uncertainty is the same methodology whether the plant is a power converter or a fly-by-wire actuator system.

Modern robust control methods — H-infinity synthesis, mu-analysis, structured singular value computation — transfer completely and are directly applied in aerospace. The flight control community has been one of the primary drivers of robust control methodology development since the 1980s, precisely because the flight control problem has the features that robust control addresses: significant model uncertainty (aerodynamic derivatives are known only to 10–20% accuracy from wind tunnel data), multiple uncertain parameters (Mach number, altitude, mass, and centre of gravity all change the plant), and stringent stability margin requirements across the entire uncertainty set. The control engineer with H-infinity or mu-synthesis experience is stepping into a domain that has specifically developed those methods.

State estimation — the Kalman filter and its nonlinear extensions (EKF, UKF, particle filter) — transfers completely to aerospace navigation and ADCS. The Kalman filter that fuses GPS and IMU measurements for a ground vehicle is essentially the same algorithm that fuses GPS, accelerometer, gyroscope, star tracker, and magnetometer measurements for a spacecraft attitude determination system. The error-state formulation that accommodates quaternion attitude kinematics in a spacecraft ADCS is a specific but learnable adaptation of the general nonlinear state estimation framework.

Optimal control — LQR, LQG, MPC, trajectory optimisation — transfers directly to aerospace trajectory guidance and orbit raising optimisation. The Pontryagin maximum principle that determines the minimum-time or minimum-fuel trajectory for a spacecraft is the same variational calculus result that governs minimum-time control of any dynamical system. The model predictive controller that manages constraints in a process plant is structurally identical to the MPC that manages airspeed, altitude, and energy constraints in an aircraft energy management system.

The Specific Gaps — What Does Not Transfer Without Work

The physical modes of aircraft dynamics — their meaning, their values, and their handling quality implications. The control engineer who has studied control theory in an electrical or chemical engineering context knows linear systems theory but not the specific dynamic modes of an aircraft. An aircraft in longitudinal motion has two modes: the short-period mode (a fast oscillation in angle of attack and pitch rate, with natural frequency typically 1–4 rad/s and damping ratio 0.3–0.7 for a large transport aircraft) and the phugoid mode (a slow exchange of kinetic and potential energy at nearly constant angle of attack, with natural frequency typically 0.05–0.1 rad/s and damping ratio 0.03–0.1). In lateral-directional motion, there are three modes: the roll subsidence (a first-order mode dominated by roll rate convergence), the Dutch roll (an oscillatory coupled yaw-roll mode, with natural frequency 0.5–1.5 rad/s and damping that must meet MIL-HDBK-1797B requirements), and the spiral mode (a slow first-order mode that may be unstable for many

conventional aircraft designs, but whose time-to-double is constrained by handling quality requirements). These modes are not abstract mathematical constructs — they have physical meaning that directly determines pilot workload, and the handling quality requirements that govern them are the result of decades of flight test research. The control engineer who does not know what a phugoid is cannot design a longitudinal autopilot that respects the separation between the phugoid and the altitude hold control loop bandwidth.

The aerodynamic stability and control derivative framework. Aircraft flight dynamics are described through a set of stability and control derivatives — the partial derivatives of the aerodynamic force and moment coefficients with respect to the motion variables and control surface deflections. Cl_α (the lift curve slope — how much lift increases per radian increase in angle of attack), Cm_α (the pitch moment derivative — the primary measure of pitch static stability), Cn_β (the yaw moment due to sideslip — the primary measure of directional stability), and their many companions define the plant that the flight control system must control. The control engineer must understand what these derivatives represent physically, how they vary with Mach number and altitude, and which ones are the primary drivers of aircraft handling quality. Without this framework, the control engineer cannot interpret the linearised aircraft model they are given to design against, cannot assess whether a proposed gain schedule will maintain stability across the flight envelope, and cannot explain why the control law becomes more aggressive at low dynamic pressure.

The regulatory and certification framework for flight control systems — DO-178C, ARP4754A, and the handling quality requirements of MIL-HDBK-1797B and CS-25. Flight control systems are the most safety-critical software on a certificated aircraft — a flight control law error that produces an uncommanded roll at low altitude can be catastrophic. Every flight control law implemented on a certificated aircraft is subject to DO-178C software development assurance at DAL A (for safety-critical control laws) or DAL B (for secondary control laws), and the control law development process — from requirements derivation through design through verification and validation — must meet the ARP4754A systems engineering process. The handling quality requirements of CS-25.143 (control and maneuverability requirements), MIL-HDBK-1797B (for military aircraft), and the JAR-25 background that shaped current CS-25 requirements are the regulatory standards against which the control law performance must be demonstrated. The control engineer who does not understand this framework will design control laws that are analytically sound but certificationally inadmissible — because the stability margins, the bandwidth limits, and the failure mode analysis that certification requires have not been addressed.

The structural mode coupling problem — aeroservoelasticity. The flight control system must control an airframe that is not a rigid body — it bends, twists, and vibrates at natural frequencies that may overlap with the bandwidth of the flight control loop. The first symmetric wing bending mode of a large transport aircraft may occur at 1–3 Hz; the first antisymmetric wing bending mode at 2–4 Hz; the first fuselage bending mode at 3–6 Hz. The flight control law, without careful structural mode filtering, can inject energy into these modes through the feedback loop — producing an aeroservoelastic instability that can lead to structural failure. The notch filter that attenuates the structural mode frequencies in the feedback path is the standard mitigation, but it introduces phase lag at frequencies below the notch that erodes the rigid-body stability margins. This aeroservoelastic design trade-off — structural mode attenuation versus rigid-body stability margin preservation — is a specific and non-trivial control engineering challenge that the flight control engineer must understand.

The spacecraft attitude kinematics — quaternion representation and singularity-free attitude control. The control engineer who has studied attitude control in an academic context using Euler angles will encounter the specific challenges of quaternion-based spacecraft attitude representation when working on real spacecraft ADCS. Quaternions represent the spacecraft attitude as a unit quaternion (four components subject to the unit-norm constraint), avoiding the gimbal lock singularity that afflicts 3-2-1 Euler angles near specific orientations. The attitude error formulation for a quaternion-based controller, the attitude kinematics in quaternion form (the quaternion differential equation driven by the angular velocity vector), and the Wahba problem formulation for determining the spacecraft attitude from multiple vector observations (star tracker measurements, magnetometer readings, sun sensor outputs) are the specific spacecraft attitude determination mathematics that ADCS engineers must know.

The real-time implementation constraints of flight critical control laws. The control engineer from an academic or simulation background has designed controllers that run on workstations with floating-point processors and essentially unlimited computational resources. Flight control computers are typically implemented in fixed-point arithmetic (to avoid the unpredictable timing of floating-point operations), with limited computational resources, hard real-time deadlines (the control law must complete execution within

the sample period, typically 12.5ms or 25ms for inner loops), and specific numerical precision requirements that must be verified by analysis and test. Anti-windup strategies for integrators that saturate against actuator limits, discretisation of continuous-time control laws using the Tustin or zero-order hold method with specific attention to the phase shift introduced at the sample frequency, and the finite word-length effects of fixed-point arithmetic on filter coefficient representation are the implementation engineering challenges that practical flight control law design requires.

By Career Stage

Recent Control Theory Graduate

The recent graduate has the strongest mathematical preparation of any transition candidate — state space, frequency domain, optimal control, and state estimation are current and active knowledge. The preparation task is entirely domain-specific: develop the aeronautical physics (the aircraft modes, the stability derivatives, the handling quality requirements), engage with the regulatory framework (CS-25, DO-178C, ARP4754A), and build at least one domain-specific control design exercise. The entry point is most commonly a control systems graduate scheme at an OEM or a flight control system supplier, or a controls engineer role at an autonomous flight or spacecraft company.

Mid-Career Control Engineer (5–15 years)

The mid-career control engineer has demonstrated ability to design, implement, and verify control systems in a real production context — hardware-in-the-loop testing, gain scheduling implementation, failure mode analysis, and production code generation. This production controls experience is specifically valuable to aerospace because it is what pure academic control theory backgrounds lack. The translation task is domain physics and regulatory: developing the aircraft or spacecraft dynamics knowledge and the certification framework vocabulary that makes the existing control engineering depth applicable in aerospace.

Senior Control Engineer / Technical Lead

The senior control engineer who has led control system development programmes, managed multi-disciplinary teams, and interfaced with customers and certification authorities on complex control problems has leadership competencies that map directly to flight control chief engineer, ADCS lead, and GNC technical lead roles. The domain physics and regulatory gaps remain, but the leadership and engineering depth are in the strongest position of any transition candidate.

Domain Mapping — Where Your Control Background Fits Best

Classical Control (PID, root locus, Bode, loop shaping) → Flight Control Autopilots, Engine Control (FADEC), Actuator Position Control Loops

Transfer function analysis, gain and phase margin assessment, PID tuning — transfer completely. The aerospace addition: aircraft longitudinal and lateral-directional dynamic modes and their handling quality requirements, the actuator rate and position limit anti-windup problem for flight control surfaces, and FADEC fuel metering valve control with temperature and pressure protection overrides.

Modern / Optimal Control (LQR, LQG, H-infinity, mu-synthesis) → Flight Control Law Design, Robust Stability Analysis, Envelope Expansion

State space design, LQR/LQG synthesis, H-infinity loop shaping, structured singular value analysis — transfer completely. The aerospace addition: the aerodynamic uncertainty model that defines the plant uncertainty for mu-analysis (Mach number and altitude variation of stability derivatives), the handling quality requirements in the coprime factorisation and H-infinity mixed sensitivity formulation, and the flight envelope expansion test methodology that verifies robust stability against the uncertainty model.

State Estimation (Kalman filter, EKF, particle filter) → Navigation, ADCS, Air Data Estimation

Filter architecture, process noise and measurement noise tuning, nonlinear extensions — transfer completely. The aerospace addition: the GPS/IMU integration architecture (error-state formulation for inertial navigation error dynamics), the spacecraft attitude determination problem (Wahba problem formulation, quaternion error state), and the air data estimation problem (angle of attack and sideslip estimation from three-axis accelerometer and gyroscope, used in advanced air data systems).

Guidance and Trajectory Optimisation (Pontryagin, dynamic programming, MPC) → Launch Vehicle Guidance, Spacecraft Orbit Raising, Autonomous Flight Path Planning

Variational calculus, indirect and direct trajectory optimisation, MPC formulation — transfer completely. The aerospace addition: the two-body gravitational dynamics of spacecraft orbit mechanics (Kepler's equa-

tion, orbital element representation, the impulsive manoeuvre model), the atmospheric entry guidance problem (the skip re-entry corridor and the controlled lift modulation guidance law), and the vehicle energy management problem for unpowered glide from high altitude.

Nonlinear Control (feedback linearisation, sliding mode, Lyapunov methods) → Rotorcraft Control, eVTOL Transition Control, Reentry Vehicle Control

Feedback linearisation, sliding mode control, Lyapunov stability analysis — transfer completely. The aerospace addition: the rotor aerodynamics nonlinearity (the blade element model, the inflow dynamics, the gyroscopic coupling between flapping and lead-lag) for rotorcraft, and the strong aerodynamic nonlinearity in the eVTOL hover-to-forward-flight transition regime where linear methods fail.

Adaptive Control (MRAC, L1 adaptive, self-tuning regulators) → Onboard Parameter Identification, Fault-Tolerant Flight Control, Learning Flight Control

Model reference adaptive control, online parameter estimation, stability analysis under parameter variation — transfer completely. The aerospace addition: the specific robustness requirements for adaptive flight control (the aircraft must remain controllable during the adaptation transient following a failure or a significant aerodynamic change), the NASA and DARPA research programme context for adaptive flight control, and the specific certification challenge of demonstrating stability bounds for an adaptive controller.

Embedded Control (real-time implementation, fixed-point arithmetic, code generation) → FADEC, Flight Control Computer, ADCS Flight Software

Real-time scheduling, fixed-point filter implementation, discretisation methods — transfer with DO-178C software assurance addition. The aerospace addition: DO-178C DAL A or B software development process for flight-critical control laws, the WCET analysis and fixed-point numerical precision analysis required for DO-178C compliance, and the Hardware-In-the-Loop (HIL) test methodology for flight control law verification.

PART II: 20 SPECIFIC PREP TASKS FOR THE TRANSITION

FOUNDATIONAL — DO THESE FIRST

1. Derive the linearised aircraft equations of motion — the six-DOF linearised state space model and its physical modes 8–10 hours

Skill: The physical system that aerospace flight control operates on — the foundational domain knowledge

Why the field cares: The first question that distinguishes a control engineer who has prepared for aerospace from one who has not is “what are the natural modes of the longitudinal dynamics of a large transport aircraft, and what are their physical interpretations?” Every aerospace flight control interview will probe this because it is the most direct test of whether the control engineer understands the physical system they are designing for. The linearised aircraft equations of motion — derived by expanding the full nonlinear six-DOF equations about a reference equilibrium condition and retaining only first-order terms — produce the longitudinal state space model with states (u, w, q, θ) — forward velocity perturbation, normal velocity perturbation (related to angle of attack), pitch rate, and pitch angle — and the lateral-directional model with states (v, p, r, ϕ) — side velocity (related to sideslip), roll rate, yaw rate, and roll angle. The eigenvalues of these linearised systems are the modes: for the longitudinal system, the short-period mode (two complex eigenvalues at approximately $-0.5 \pm 2j$ rad/s for a typical transport) and the phugoid mode (two complex eigenvalues at approximately $-0.02 \pm 0.1j$ rad/s). The control engineer who can derive these, interpret them physically, and relate them to the handling quality requirements is demonstrating the aerospace domain knowledge that every flight control interview assesses.

Done signal: You can write down the linearised longitudinal state space model in the standard form $\dot{x} = Ax + Bu, y = Cx$, identifying the meaning of each state variable, each element of the A matrix (the aerodynamic and gravitational derivatives — $X_u, Z_u, M_u, Z_\alpha, M_\alpha, M_q$, and others), and each element of the B matrix (the elevator control effectiveness derivatives Z_{δ_e} and M_{δ_e}). You can compute

8–10 hours

the eigenvalues of the A matrix for a specified set of aerodynamic derivatives, identify which eigenvalue pair corresponds to the short-period and which to the phugoid, explain the physical motion of each mode (short-period: rapid oscillation in α and q with nearly constant speed; phugoid: slow exchange of speed and altitude with nearly constant α), and state the typical natural frequency and damping ratio ranges for each mode in a large transport aircraft.

6–8 hours

2. Understand the aircraft stability and control derivative framework — what each derivative means physically and how it varies with flight condition

Skill: The physical aerodynamic parameters that define the plant the flight control system must control

Why the field cares: The aerodynamic stability and control derivatives are the coefficients of the linearised aircraft equations of motion — they translate aerodynamic forces and moments into the mathematical terms of the state space model. Understanding what each derivative means physically — not just as a matrix entry but as an aerodynamic phenomenon — is the domain knowledge that allows the control engineer to interpret gain schedules, understand why the plant becomes more difficult to control at certain flight conditions, and assess whether a proposed control law will remain stable as the derivatives vary. $C_{l\alpha}$ is the lift curve slope — approximately 2π per radian for a thin wing in subsonic incompressible flow, reduced by planform effects and increased by compressibility effects; it increases with Mach number up to the transonic regime and then drops sharply through the shock formation process. $C_{m\alpha}$ is the pitch stiffness derivative — negative for a statically stable aircraft (an angle of attack increase produces a nose-down pitching moment that restores the aircraft to equilibrium); it becomes less negative (and may become positive — statically unstable) at high Mach numbers for many configurations. $C_{n\beta}$ is the directional stability derivative — positive for a directionally stable aircraft (sideslip produces a restoring yawing moment toward zero sideslip); it may become negative for highly swept delta wings at high angle of attack.

Done signal: For each of the ten most important longitudinal and lateral-directional stability and control derivatives ($C_{l\alpha}$, $C_{m\alpha}$, C_{mq} , $C_{Z\delta_e}$, $C_{m\delta_e}$, $C_{y\beta}$, $C_{l\beta}$, $C_{n\beta}$, $C_{l\delta_a}$, $C_{n\delta_r}$), you can state: the physical meaning (what aerodynamic phenomenon it measures), the sign for a conventionally stable aircraft, the typical numerical magnitude range for a large transport aircraft, and how the derivative changes with Mach number and angle of attack at a qualitative level. You can explain why $C_{m\alpha}$ becoming less negative at high Mach number requires the flight control system to increase the pitch control authority at high speed to maintain the same handling quality, and you can connect this requirement to gain scheduling of the pitch control law.

6–7 hours

3. Understand the handling quality requirements — MIL-HDBK-1797B and CS-25 — and their control design implications

Skill: The performance specifications that the flight control law must meet — the control design requirements in their regulatory form

Why the field cares: The handling quality requirements — the standards that quantify how the aircraft must respond to pilot inputs and atmospheric disturbances — are the control system specifications that the flight control law designer must satisfy. MIL-HDBK-1797B (Military Specification — Flying Qualities of Piloted Vehicles) defines three levels of handling quality: Level 1 (adequate for the mission without increasing pilot workload), Level 2 (adequate but increased pilot compensation required), and Level 3 (inadequate but aircraft can be safely recovered). Each level is quantified through specific requirements on the modal parameters: the short-period natural frequency and damping ratio must fall within specified Level 1 boundaries that depend on the flight phase and the aircraft's pitch sensitivity (the ratio n/α — normal acceleration per unit angle of attack); the Dutch roll damping ratio must be above 0.04 (Level 1) or 0.02 (Level 2); the roll mode time constant must be below specified values; and the bandwidth and phase delay requirements of ADS-33 (for rotorcraft) provide an alternative formulation for bandwidth-based control law assessment. CS-25.143 (Control and Maneuverability) provides the civil aircraft equivalent requirements — less quantitatively

6–7 hours

prescriptive than MIL-HDBK-1797B but establishing the framework within which handling quality must be demonstrated.

Done signal: You can describe the MIL-HDBK-1797B short-period requirements — the Level 1, 2, and 3 natural frequency and damping ratio boundaries for Category B flight phases (normal operation) — and explain what these requirements mean for the flight control law design (the closed-loop system with the flight control law active must place the short-period mode within the Level 1 boundary). You can explain what the C^* criterion is — the linear combination of pitch rate and normal acceleration ($C^* = q + V_T/g \times n_z$) that provides a consistent response measure across speeds — and describe why it is used as the primary command variable in pitch control laws for transport aircraft (it produces consistent feel forces and response characteristics independent of airspeed). You can describe what bandwidth and phase delay mean for handling quality assessment (bandwidth: the frequency at which the attitude response phase drops to -135° or the gain drops 6 dB; phase delay: the additional phase lag beyond 180° at twice the bandwidth frequency, divided by 2 — a measure of the high-frequency phase roll-off that correlates with pilot-induced oscillation susceptibility).

5–6 hours

4. Understand the structural mode interaction problem — aeroservoelasticity and the notch filter trade-off

Skill: The coupling between flexible structure and flight control that is the most distinctive challenge in aerospace flight control

Why the field cares: The aeroservoelastic (ASE) problem is the most specifically aerospace of all the challenges in flight control system design. An aircraft is not a rigid body: the wings, fuselage, and control surfaces flex and vibrate at natural frequencies that depend on the structural stiffness and mass distribution. If the flight control system has sufficient gain at these structural frequencies — which it will unless specifically mitigated, because the flight control bandwidth often extends to frequencies approaching the first structural mode — the closed-loop system can pump energy into the structural modes, driving oscillation that can lead to structural failure. The notch filter in the feedback path attenuates the structural mode frequency in the sensor signal before it re-enters the control loop, preventing the destabilising feedback. But the notch filter introduces phase lag at frequencies below the notch, reducing the phase margin of the rigid-body control loop. The design trade-off — notch depth (more depth = better structural mode attenuation but more rigid-body phase lag) versus notch width (broader notch = more robust to structural mode frequency uncertainty but more rigid-body phase lag) — is the specific ASE design challenge that the flight control engineer faces and that aerospace interviews probe specifically.

Done signal: You can explain what aeroservoelastic instability is — the closed-loop instability that arises when the flight control system provides sufficient feedback gain at the structural mode frequencies to sustain oscillation, through the mechanism of the aerodynamic loads on the flexed surface feeding back through the control law — and describe the physical consequence of ASE instability onset (rapid structural oscillation that can lead to failure within seconds). You can describe the notch filter design trade-off: a deeper notch and narrower bandwidth provides better structural mode attenuation but introduces more phase lag at the rigid-body frequencies; a shallower notch and wider bandwidth provides less phase lag but requires larger structural mode frequency margins. You can explain why the structural mode frequency uncertainty — the difference between ground vibration test results and in-flight structural mode frequencies, caused by aerodynamic loading on the structure — drives the notch filter to be wider than the ground test mode frequency alone would require.

6–8 hours

5. Understand spacecraft attitude kinematics — quaternion representation, Euler's equation, and the attitude control problem

Skill: The spacecraft domain equivalent of aircraft flight dynamics — essential for ADCS roles

Why the field cares: Spacecraft attitude control is one of the most directly accessible aerospace control applications for the control theory specialist, because the physical dynamics are relatively simple (Euler's equation for rigid body rotation) and the control problem is well-posed (regulate attitude

6–8 hours

to a reference with specified accuracy and bandwidth). The specific mathematical tools required: quaternion kinematics (the quaternion q represents the rotation from the inertial to the body frame; the quaternion kinematic equation $\dot{q} = \frac{1}{2}\Xi(q)\omega$ relates the quaternion time derivative to the body angular velocity vector ω), Euler's equation for rigid body rotation ($I\dot{\omega} + \omega \times I\omega = \tau$, where I is the inertia tensor and τ is the applied torque vector), and the attitude error quaternion formulation for the control law (the error quaternion $q_e = q_d \otimes q^{-1}$, the product of the desired quaternion and the inverse of the current quaternion, represents the rotation needed to align the current attitude with the desired attitude). The actuators for attitude control — reaction wheels, magnetorquers, and thrusters — each have specific momentum capabilities and constraints that determine the achievable attitude control bandwidth.

Done signal: You can write the quaternion kinematic equation and the Euler rigid body rotation equation, and combine them to form the full spacecraft attitude dynamics model in state space form with state vector $[q, \omega]$. You can formulate the quaternion error attitude control law for a reaction wheel-actuated spacecraft — the PD controller that maps the attitude error (the vector component of the error quaternion) and the angular velocity error to a commanded reaction wheel torque — and assess the linearised closed-loop stability by computing the eigenvalues of the linearised error dynamics. You can explain what a reaction wheel saturation event is — when the wheel angular momentum reaches the maximum rated value and can no longer provide additional control torque in the saturation direction — and describe the desaturation strategy (using thrusters or magnetorquers to dump the excess angular momentum from the wheel to the environment).

5–6 hours

6. Understand the DO-178C and ARP4754A implications for flight control software — the regulatory framework for safety-critical control laws

Skill: The software development assurance and systems engineering process that governs every certified flight control law

Why the field cares: The flight control law that is analytically correct — that achieves the required handling quality, robustness, and stability margins — is not a certified flight control law until it has been developed through a process that meets DO-178C (for the software implementation) and ARP4754A (for the system-level development assurance). For a flight-critical control law at DAL A: the control law requirements must be formally documented before any code is written, the code must be verified against every requirement by an independent verifier, structural coverage at the MC/DC level must be demonstrated for every code path, and the software baseline must be under formal configuration management. The specific flight control law development considerations: the discretisation of a continuous-time control law (the process of converting the continuous-time transfer function to a discrete-time equivalent using the Tustin or ZOH method) must be documented and its accuracy verified by analysis; the fixed-point implementation of the discrete filter (the conversion of floating-point coefficients to fixed-point representation and the verification that the resulting filter meets the performance specification despite finite word-length effects) must be specifically analysed and documented; and the anti-windup implementation for integrators that saturate against actuator limits must be verified to prevent integrator saturation during envelope exceedance.

Done signal: You can explain the DO-178C requirement that applies to a flight-critical control law (DAL A: all objectives of DO-178C Table A-1 through A-10 apply, including independent review, MC/DC structural coverage, and formal CM). You can describe the specific implementation verification steps required for a discretised control law: the comparison of the discrete-time transfer function Bode plot against the continuous-time design (to verify that the discretisation method has not introduced significant phase error at the control loop crossover frequency), and the fixed-point simulation that compares the fixed-point implementation against the floating-point reference (to verify that the truncation and rounding errors do not degrade the closed-loop stability margin below the specification minimum). You can explain what anti-windup is and why it is necessary for integral states in a flight control law (without anti-windup, the integrator continues to accumulate when the actuator saturates, producing a large integrator state that causes the actuator to remain at the saturation limit long after the command has returned to the linear range, resulting in an overshooting transient called integrator windup).

5–6 hours

7. Understand gain scheduling — the methods for handling nonlinear flight envelope variation in a linear control law

Skill: The primary practical method for extending linear control law designs across the nonlinear flight envelope

Why the field cares: Linear control law design produces a controller that is optimal at the design point — a specific Mach number, altitude, and configuration. As the aircraft moves away from the design point, the aerodynamic derivatives change: the control effectiveness ($C_{m\delta_e}$) increases at higher dynamic pressure, the pitch stiffness ($C_{m\alpha}$) changes with Mach number, and the stability margins of the design-point controller degrade. Gain scheduling — the systematic variation of controller gains with the measured flight condition (typically dynamic pressure, Mach number, and angle of attack) — is the standard method for maintaining acceptable control law performance across the flight envelope. The gain schedule design process: design linear controllers at a grid of flight conditions (the design points), verify that each design meets the handling quality requirements at its design point, interpolate the gains between design points, and verify that the interpolated gains maintain stability and performance at all intermediate conditions. The interpolation method — linear interpolation versus spline interpolation versus linear parameter-varying (LPV) scheduling — affects the stability of the gain-scheduled controller at conditions between design points, because a gain-scheduled controller is not automatically stable between design points even if it is stable at each design point.

Done signal: You can describe the gain scheduling design process for an aircraft pitch control law — the grid of design points in the Mach-altitude space, the linear controller designed at each grid point, the gain table that stores the scheduled gains, and the interpolation logic. You can explain what frozen-point stability analysis means — computing the eigenvalues of the closed-loop system with gains fixed at the scheduled values for a specific flight condition — and why it is a necessary but not sufficient condition for stability of the gain-scheduled system. You can describe one specific gain schedule variable for a pitch control law and explain the physical reason it is used — for example, dynamic pressure ($\bar{q}$) schedules the pitch rate gain because the pitch moment effectiveness is proportional to $\bar{q}$, so dividing by $\bar{q}$ (or scheduling the gain inversely with $\bar{q}$) maintains constant closed-loop pitch stiffness across speeds.

5–6 hours

8. Understand the FADEC — Full Authority Digital Engine Control — and its control architecture

Skill: The aircraft engine control system — a complex multi-loop control problem with specific safety and performance requirements

Why the field cares: The Full Authority Digital Engine Control is the embedded control system that governs every aspect of a turbofan engine's operation: the fuel metering valve position that controls thrust, the variable stator vane schedules that optimise compressor efficiency and surge margin, the turbine clearance control that minimises tip clearance for efficiency, and the engine protection logic that prevents over-temperature, over-speed, and compressor surge. The FADEC is a multi-loop control system with nested loops of different bandwidth: the N2 (high-pressure spool speed) control loop at the outermost level, the fuel flow control loop at an intermediate level, and the fuel metering valve position loop at the innermost level. Engine acceleration and deceleration control is a particularly complex challenge: the engine must accelerate from idle to maximum thrust as rapidly as possible (to respond to a go-around or missed approach command) without exceeding the surge boundary of the compressor (which would cause an uncontrolled loss of compression and potentially engine damage). The surge boundary is typically scheduled as a function of N2 and the inlet total temperature, and the FADEC acceleration schedule must ensure the operating point remains below the surge boundary throughout the acceleration.

Done signal: You can describe the FADEC control architecture for a turbofan engine — the primary control variable (N2 or EPR — Engine Pressure Ratio), the fuel metering valve as the primary actuator, the variable geometry actuators (VSVs, VBV — Variable Stator Vanes, Variable Bleed Valves) as secondary actuators, and the protection logic overlays (T4 — turbine inlet temperature limit, N1 and N2 speed limits, surge recovery). You can explain what compressor surge is — the aerodynamic instability that occurs when the compressor operating point crosses the surge line, characterised by a violent reversal of the flow through the compressor — and describe the margin between the normal

5–6 hours

operating line and the surge boundary that the FADEC acceleration schedule must maintain. You can describe what the throttle response requirement under CS-E 650 specifies — the maximum time from idle to takeoff thrust, typically 8 seconds for large turbofan engines — and explain the control engineering challenge of meeting this requirement while maintaining surge margin.

2–3 hours

9. Prepare specific control engineering stories with aerospace context translation

Skill: The interview narrative that demonstrates control engineering depth with aerospace domain awareness

Why the field cares: Aerospace control system interviews probe the specific control engineering problems the candidate has solved — the specific instability diagnosed, the specific robust stability analysis performed, the specific implementation problem resolved. The control engineer's strongest interview stories are specific control problems: a process control loop that was exhibiting limit cycling (the diagnosis of the integrator windup in the primary flow controller, the anti-windup modification, and the loop gain reduction that eliminated the cycle), a servo system that was marginally stable at low temperature (the analysis of the gain increase caused by reduced fluid viscosity in the hydraulic actuator and the gain schedule modification that maintained stability across temperature), or a state estimator that was diverging due to an unmodelled bias in the accelerometer (the EKF model extension to include accelerometer bias as an additional state and the improvement in filter convergence). The aerospace translation that accompanies each story — “in a flight control system, the equivalent investigation would be” — demonstrates the specific domain awareness that distinguishes a prepared candidate.

Done signal: You have three genuine control engineering stories from your career, each with: the specific system (with the plant transfer function or state space model if known), the specific control problem (instability, limit cycle, poor disturbance rejection, diverging estimator), the analysis tools used (Bode plot, root locus, eigenvalue analysis, Monte Carlo simulation), the root cause identified, the control law modification implemented, the verified improvement, and the aerospace translation. At least two stories involve closed-loop stability issues rather than reference tracking performance, because stability is the primary concern in aerospace control.

30–60 hours

10. Build a demonstrable aerospace control domain engagement — a flight control or ADCS design exercise

Skill: The portfolio project that converts control theory expertise into aerospace domain demonstration

Why the field cares: The control engineer who arrives at an aerospace interview with a documented flight control or ADCS design exercise — not merely described interest in aerospace, but specific control design work in an aerospace context — is immediately differentiated. Accessible portfolio projects: design a full longitudinal flight control law for the NASA Generic Transport Model (GTM) — a publicly available linearised aircraft model with documented aerodynamic derivatives — including a pitch rate command inner loop, an altitude hold outer loop, and gain scheduling with dynamic pressure; the GTM is available from the NASA Langley MATLAB/Simulink toolbox and has been used in hundreds of academic flight control studies; design a quaternion-based PD attitude controller for a small satellite (1U or 3U CubeSat) with reaction wheel actuation, including attitude determination from sun sensor and magnetometer measurements using the TRIAD method, and verify the closed-loop performance against a pointing requirement of 5° accuracy; or design an EKF-based GPS/IMU navigation filter for a fixed-wing UAV, using the standard error-state formulation, calibrate against the standard flight test manoeuvre set, and verify the filter against simulated GPS outage. Each of these is achievable in 30–60 hours using MATLAB/Simulink (standard tool for aerospace control) and produces specific, documented output that demonstrates aerospace domain engagement at technical depth.

Done signal: You have completed at least one portfolio project in an aerospace control domain that produces a documented control law design and verification — a MATLAB/Simulink model,

30–60 hours

a Jupyter notebook, or equivalent — including the plant model (described with its aerodynamic or attitude dynamics parameters), the control law design methodology (the design objectives, the loop shaping approach, the gain schedule or Kalman filter design), the stability margin verification (Bode plots showing gain and phase margins, or eigenvalue analysis), and a simulation of the closed-loop response. The documentation explicitly references the relevant aerospace requirement (handling quality requirement, pointing accuracy requirement, navigation accuracy requirement) and demonstrates that the design meets or misses it by a quantified amount.

ROLE-SPECIFIC — TARGET AFTER FOUNDATIONAL TASKS

5–6 hours

11. For flight control roles: understand the fly-by-wire architecture — the sensor suite, the flight control computer, and the actuator control electronics

Skill: The hardware architecture within which flight control laws execute

Why the field cares: The flight control law exists within a hardware architecture — the flight control computer (FCC), the inertial reference system (IRS) or air data inertial reference unit (ADIRU), the actuator control electronics (ACE), and the primary and secondary flight control actuators — that determines how the control law is implemented, what the latency is from sensor measurement to actuator command, and what the failure modes are that the control law must accommodate. The FCC receives sensor signals (angle of attack from air data sensors, angular rates from the IRS, normal acceleration from accelerometers, control surface position from position transducers), executes the control law, and outputs actuator commands to the ACE. The ACE implements an inner actuator position control loop (typically at 80Hz or higher) that converts the FCC's force or position command to the actuator servo-valve current. The total latency from sensor measurement to actuator response — the sum of the sensor signal conditioning delay, the FCC execution frame time, the ACE execution frame time, and the actuator hydraulic response time — is typically 60–120ms for a modern fly-by-wire aircraft, and this latency must be accounted for in the flight control law phase margin assessment.

Done signal: You can describe the signal path from a pilot pitch command input to the elevator deflection in a fly-by-wire aircraft — the sidestick or control column position signal through the FCC input, the control law computation, the elevator position command transmission to the ACE, the ACE servo loop that drives the electrohydraulic servo-valve, and the actuator cylinder that moves the elevator. You can estimate the total latency in this signal path (approximately 80–100ms for a typical modern FBW aircraft) and explain how this latency is represented in the control law frequency domain analysis (a dead time approximation e^{-Ls} in the loop transfer function, or a Padé approximation for root locus analysis). You can describe what a triplex or quadruplex voter does in a redundant FCC architecture — comparing the outputs of three or four independent computers and using majority vote or cross-channel monitoring to detect and isolate a failing computer.

5–6 hours

12. For ADCS roles: understand the Wahba problem and attitude determination from multiple vector measurements

Skill: The attitude determination algorithm that fuses multiple sensor measurements into a quaternion estimate

Why the field cares: The spacecraft attitude must be determined from sensor measurements before it can be controlled — this is the attitude determination problem. If two or more reference vectors are known in both the inertial frame (from an ephemeris model or direct measurement) and the body frame (from sensor measurements), the spacecraft attitude can be determined uniquely. The Wahba problem is the optimal estimation formulation: given n reference vectors $\{r_i\}$ known in the inertial frame and their measured counterparts $\{b_i\}$ in the body frame, find the rotation matrix A that minimises the loss function $L(A) = \frac{1}{2} \sum_i w_i |b_i - Ar_i|^2$. The q-method, QUEST, TRIAD, and SVD methods are the standard algorithms for solving the Wahba problem. In practice, the spacecraft ADCS uses a star tracker (which measures the directions to multiple stars simultaneously, providing

5–6 hours

high-accuracy attitude determination to better than 1 arcsecond), a magnetometer (which measures the Earth's magnetic field direction, providing coarser attitude information), and sun sensors (which measure the sun direction, useful when the star tracker is blinded by the sun or Earth limb).

Done signal: You can describe the TRIAD method for attitude determination from two reference vectors — the construction of two orthogonal frames (one from the primary reference vector, one from the cross product of the primary and secondary reference vectors) in both the inertial and body frames, and the computation of the rotation matrix from the two frames as the product of the body frame matrix and the transpose of the inertial frame matrix. You can explain why the TRIAD method is suboptimal (it ignores the measurement noise statistics and treats the two vector measurements unequally, using the primary vector to define one axis exactly regardless of its measurement accuracy) and describe what the QUEST algorithm adds (it incorporates measurement noise covariances as weights in the Wahba loss function, producing the minimum-variance attitude estimate for the given measurement accuracies). You can describe what a star tracker provides as an output — typically a quaternion and an angular velocity estimate, derived from successive frame-to-frame tracking of star centroids — and explain why it is the most accurate attitude determination sensor currently in use.

6–8 hours

13. For GNC roles: understand the navigation filter — the GPS/IMU integration architecture and the error-state formulation

Skill: The navigation state estimator that is the most widely deployed Kalman filter application in aerospace

Why the field cares: The GPS/IMU integrated navigation filter is the workhorse state estimator of aerospace guidance, navigation, and control. The IMU (Inertial Measurement Unit — an accelerometer triad and a gyroscope triad) measures specific force and angular velocity continuously at high rate (typically 100–1000 Hz), but accumulates position and attitude errors over time due to sensor bias, scale factor errors, and random walk. GPS provides absolute position and velocity measurements at lower rate (typically 1–10 Hz) with bounded errors, but is subject to outages, multipath, and jamming. The integrated filter exploits the complementary error characteristics: the IMU's high-rate, relative measurement is used to propagate the navigation state between GPS updates, while the GPS measurement is used to correct the accumulated IMU errors. The error-state formulation of the Kalman filter — which estimates the error in the navigation state (position error, velocity error, attitude error, IMU bias) rather than the navigation state itself — is the standard architecture for GPS/IMU integration because it allows the attitude error to be represented as a small rotation (three-component error vector) rather than requiring the full nonlinear quaternion kinematics in the filter.

Done signal: You can describe the error-state GPS/IMU integration filter — the three-component position error state, the three-component velocity error state, the three-component attitude error state (small angle approximation in the locally level frame), the three-component accelerometer bias state, and the three-component gyroscope bias state — and write the linearised error dynamics matrix (the F matrix) for this 15-state system. You can describe the GPS measurement model in terms of the error state — the position measurement residual (GPS position minus IMU position) as a linear function of the position error state — and the Kalman gain computation. You can explain what happens during a GPS outage — the IMU error states grow according to the error dynamics model, position error accumulates as a double integral of accelerometer bias, and the filter covariance matrix expands to reflect the growing navigation uncertainty.

5–6 hours

14. For autonomous flight roles: understand the trajectory planning and control architecture for UAVs and eVTOL

Skill: The guidance and control architecture for autonomous aircraft — the fastest-growing aerospace control domain

Why the field cares: Autonomous aircraft — UAVs, UAS, and eVTOL air taxis — require a hierar-

5–6 hours

chical guidance and control architecture that extends from mission-level path planning down to the inner loop attitude control. The architecture has three levels: the path planner (which generates a feasible trajectory from the current position to the goal, typically using A*, RRT, or a polynomial trajectory method, subject to obstacle avoidance and airspace constraints), the trajectory tracker (which generates acceleration commands that follow the planned trajectory, typically using a nonlinear or linear model-predictive controller with attitude and position as states), and the attitude controller (which tracks the acceleration commands by commanding the rotor speeds or control surface deflections). For a multirotor eVTOL, the inner loop attitude controller is a proportional-derivative (PD) or LQR controller on roll, pitch, and yaw with the four motor speeds as actuators (for a quadrotor); the motor speed commands are computed from the desired roll, pitch, and yaw moments plus the thrust command through the motor mixing matrix. Understanding this hierarchical architecture — and specifically the decoupling assumptions that allow the three levels to be designed independently — is the autonomous flight control knowledge that eVTOL and UAV companies require.

Done signal: You can describe the quadrotor control architecture — the outer loop position PD controller that generates desired roll, pitch, and yaw from position error, the inner loop attitude PD or LQR controller that generates desired rotor speed increments from attitude error, and the motor mixing matrix that converts the total thrust and torque demands to individual motor speed commands. You can derive the linearised quadrotor dynamics model (the small-angle approximation that decouples the longitudinal (x-z-pitch) and lateral (y-z-roll) dynamics from the yaw dynamics) and compute the eigenvalues to assess the open-loop stability (the quadrotor is unstable in pitch and roll without control). You can explain what the trajectory tracking problem requires — the minimum snap trajectory method that generates position trajectories with bounded jerk, making the attitude commands smooth enough to be tracked by the inner attitude loop without exciting the structural resonances of the vehicle.

5–6 hours

15. For launch vehicle guidance roles: understand the powered explicit guidance algorithm and the thrust vector control problem

Skill: The guidance and attitude control of launch vehicles during ascent

Why the field cares: Launch vehicle guidance and control is one of the most demanding aerospace control problems: the vehicle is undergoing large, rapid changes in mass (as propellant is consumed), aerodynamic environment (from subsonic through transonic through supersonic to vacuum), structural loading (aerodynamic bending moment at maximum dynamic pressure), and guidance requirements (from the vertical rise phase through the pitch-over and gravity turn to the target orbit injection condition). The powered explicit guidance (PEG) algorithm — the standard terminal guidance law for orbit insertion — computes the thrust direction and duration required to achieve the desired final orbit state from the current state, using an analytical targeting solution that accounts for the vehicle's propulsion parameters and the current position and velocity. The thrust vector control (TVC) system — the gimballed engine or vernier thruster system that controls the direction of the main thrust vector — is the attitude control actuator for launch vehicles, and its control law must maintain attitude stability against the bending moment loads and the sloshing of liquid propellant in the tanks.

Done signal: You can describe the powered explicit guidance algorithm at a conceptual level — the targeting of a specific orbit energy, inclination, and right ascension of the ascending node from the current position-velocity state; the computation of the required burn direction and duration; and the iteration of the algorithm each guidance cycle as the vehicle state evolves. You can explain why PEG is called “explicit” (it generates an explicit analytic solution for the burn duration and direction, rather than iteratively optimising a trajectory in real time). You can describe the propellant slosh problem — the low-frequency oscillation of liquid propellant in a partially-filled tank, with a natural frequency that decreases as the tank empties — and explain why it is specifically challenging for the attitude control law (it adds a lightly-damped mode at a frequency near the TVC bandwidth, creating a potential resonance that must be specifically managed through notch filtering or propellant slosh suppression).

4–5 hours

16. Understand the flight simulation environment — the role of the simulator in flight control law development and verification**Skill:** The primary development and verification tool for flight control laws

Why the field cares: Flight control law development is almost entirely done in simulation before any flight test is performed — not because simulation is a substitute for flight test, but because the simulation environment allows exhaustive testing of the control law across the full envelope, across all failure conditions, and with pilots in the loop, at a cost and schedule that flight test cannot approach. The flight simulator for control law development is a high-fidelity nonlinear aircraft model (a six-DOF rigid body model with detailed aerodynamic tables, propulsion model, actuator dynamics model, sensor model, and atmosphere model) implemented in MATLAB/Simulink, FORTRAN, or C++, typically running faster than real-time on a simulation cluster for batch testing and at real-time for pilot-in-the-loop evaluation on a fixed-base or motion-base simulator. The control law verification using the simulator covers: robustness evaluation, handling quality evaluation (pilot-in-the-loop simulation with evaluation pilots rating the aircraft against the Cooper-Harper scale), failure case evaluation (the control law response to each credible failure mode, including sensor failures, actuator failures, and computer failures), and envelope protection evaluation.

Done signal: You can describe the Monte Carlo robustness simulation process for flight control law verification — the generation of a set of perturbed aerodynamic derivative combinations (typically by sampling from a uniform or Gaussian distribution centred on the nominal values with ± 10 – 20% uncertainty bounds), the execution of the control law simulation at each perturbed plant, the extraction of the relevant stability metric (gain margin, phase margin, or closed-loop pole location) at each perturbed plant, and the statistical summary that demonstrates the required probability of meeting the specification. You can explain what a Cooper-Harper rating is — the pilot workload scale from 1 (best) to 10 (worst) used to rate the handling quality of an aircraft in a specific mission task — and describe at what Cooper-Harper rating threshold the Level 1 handling quality boundary is defined (typically $CH \leq 3.5$ for Level 1).

5–6 hours

17. Understand aeroelastic flutter — what it is, what causes it, and why it constrains the flight envelope**Skill:** The aeroelastic instability that constrains the flight envelope and interacts with the flight control system

Why the field cares: Flutter is the dynamic aeroelastic instability in which the aerodynamic forces on a flexible structure extract energy from the flow and feed it into the structural oscillation — causing the oscillation amplitude to grow exponentially until structural failure occurs. It arises from the coupling of two structural modes (typically the bending and torsion modes of the wing) through the aerodynamic forces: the bending oscillation changes the wing angle of attack, producing aerodynamic pitching moment that drives the torsion mode, which produces lift variation that feeds the bending mode. At the flutter speed (the airspeed at which the coupled bending-torsion oscillation becomes neutrally stable), the energy input from the aerodynamic coupling exactly balances the structural damping — above the flutter speed, the oscillation grows without bound. The flight envelope is certified with a flutter clearance margin: the demonstrated flutter-free envelope must extend to $1.15 \times$ the aircraft's maximum operating speed (VD) under CS-25.629. The flight control system interacts with flutter in two ways: it can suppress flutter through active flutter suppression, and it can cause flutter through aeroservoelasticity.

Done signal: You can explain the mechanism of classical binary flutter — the coupling of wing bending and torsion modes through the aerodynamic moment that each mode's motion generates on the other — and describe how the flutter speed depends on the structural natural frequencies, the structural coupling, and the aerodynamic derivative. You can explain what the V-g diagram is — the plot of structural mode damping (g) and frequency versus airspeed for the flutter mode — and identify the flutter speed as the airspeed at which the damping crosses zero. You can describe what active flutter suppression does — how outboard aileron deflections can modify the wing aerodynamic load distribution to reduce the flutter mode aerodynamic excitation — and identify what the control law bandwidth requirement is for an active flutter suppression system to be effective (the control band-

width must be above the flutter mode frequency, which may be 5–15 Hz for typical transport aircraft).

5–6 hours

18. Build quantitative estimation fluency for aerospace control problems — stability margins, mode characteristics, filter requirements

4–5 hours

Skill: First-principles rapid assessment of aerospace control system performance

Why the field cares: Senior aerospace control engineers are expected to assess the plausibility of control law performance claims without running a full simulation — to estimate whether a proposed control law can achieve the required stability margin at the flight condition with the most adverse aerodynamics, to assess whether the notch filter centre frequency is correctly placed for the structural mode being attenuated, or to estimate the Kalman filter state error covariance at steady state from the sensor noise and process noise parameters. These estimates are tractable from control theory fundamentals applied to the aerospace domain: computing the crossover frequency and phase margin from the loop transfer function Bode plot, estimating the notch filter bandwidth from the structural mode frequency uncertainty and the required rigid-body phase margin preservation, or computing the steady-state Kalman gain from the algebraic Riccati equation solution for a simple second-order system. None of these requires a simulation — they require the governing physical relationships and engineering judgment about typical parameter values.

Done signal: You have worked through five aerospace control estimation problems covering at least two of: phase margin from the loop transfer function Bode plot at a specified crossover frequency with a dead time of 80ms; notch filter bandwidth required to maintain 45° rigid-body phase margin while attenuating a structural mode at 3 Hz by 20 dB; LQR gain matrix for a simplified second-order pitch rate control system with a specified state weighting; EKF steady-state error covariance for a simple attitude error estimation problem with specified gyroscope and star tracker noise; and the minimum time to double for a spiral mode given a lateral-directional state space model. You state assumptions before numbers and identify the most load-bearing approximation in each.

POSITIONING AND PRESENTATION

19. Prepare a specific answer to "Why aerospace — and what from your control background makes you ready for this role?"

2–3 hours

Skill: The transition motivation that demonstrates both control engineering depth and aerospace domain preparation

Why the field cares: The aerospace flight control hiring manager's concern with a non-aerospace control engineer is whether they understand the physical domain — specifically whether they know what a phugoid mode is, what a Dutch roll requirement means, and why the structural mode creates a stability problem — rather than just the mathematics. The answer "I have 8 years of robust control design experience and I am very interested in flight control" is not sufficient. The answer "I have spent 8 years of H-infinity and mu-synthesis experience in industrial servo-system design. I have specifically studied the aircraft flight dynamics context: I know the short-period and phugoid modes, I have designed a pitch rate command flight control law for the NASA GTM model achieving 8 dB gain margin and 45° phase margin across the flight envelope, and I understand the structural mode notch filter design trade-off — I chose a 4 Hz notch bandwidth to maintain 40° rigid-body phase margin while providing at least 20 dB attenuation at the first wing bending frequency. The specific robustness analysis I want to develop is the mu-synthesis approach to the gain schedule uncertainty problem, which my experience in servo-system robust control directly prepares me for" is demonstrating the mathematical depth and aerospace domain preparation that the flight control interviewer is assessing.

Done signal: You have a 2-minute answer that: names the specific aerospace control domain you are targeting (flight control, ADCS, GNC, FADEC, autonomous flight, or launch vehicle guidance); connects your primary control methodology (classical PID, modern state space, H-infinity, adaptive, or model predictive) to a specific aerospace control problem where that methodology is specifically

applicable; demonstrates that you have studied the aerospace-specific physics (naming the specific aircraft mode, spacecraft kinematics, or engine dynamic that your control law must address); and references your portfolio project as demonstrable evidence of the aerospace domain engagement.

2–3 hours

20. Map your control specialisation to specific aerospace employers and roles — and understand the simulation tool expectations

4–6 hours

Skill: The targeted application strategy that focuses preparation on the highest-probability opportunities

Why the field cares: The aerospace control engineering job market has specific tool expectations that differ by employer and application domain. Flight control law design at Airbus and Boeing is done primarily in MATLAB/Simulink with the Aerospace Toolbox and the Robust Control Toolbox; ADCS design at most spacecraft companies uses MATLAB/Simulink with a custom dynamics model; autonomous flight control at eVTOL companies uses a mixture of ROS (Robot Operating System), Python, and MATLAB/Simulink; FADEC control at engine companies uses a combination of MATLAB and proprietary simulation tools. The control engineer who lists MATLAB/Simulink as their primary tool is well-positioned for the majority of aerospace control engineering roles. The control engineer who has only used Python (scipy.signal, control library) or Modelica has a tool gap that some employers will accept (willingness to learn MATLAB is typically sufficient) but that should be explicitly addressed. Additionally, the specific application domain knowledge matters for employer targeting.

Done signal: You have identified at least five specific employers in your target aerospace control domain — distinguishing between OEM flight control groups, flight control system suppliers, spacecraft ADCS specialists, autonomous flight companies, and engine control specialists — and have verified that your control specialisation and tool background is specifically relevant to each. You know whether MATLAB/Simulink or Python is the primary design tool at each employer, whether DO-178C code generation (from Simulink Embedded Coder or similar) is required, and what the typical interview assessment format is. You have prepared applications for at least two employers, with a CV that explicitly maps your control engineering specialisation to the aerospace control problem domain and a cover letter that demonstrates knowledge of the specific control challenges the employer is working on.

SUMMARY REFERENCE TABLE

#	Task Description
1 All / Do first	Linearised aircraft equations of motion — six-DOF linearisation, short-period, phugoid
2 All	Aerodynamic stability and control derivatives — physical meaning, values, envelope variation
3 Flight control	Handling quality requirements — MIL-HDBK-1797B, CS-25.143, C* criterion, bandwidth
4 Flight control	Aeroservoelasticity — structural mode coupling, notch filter trade-off, V-g diagram
5 ADCS roles	Spacecraft attitude kinematics — quaternion representation, Euler equation, ADCS architecture
6 All	DO-178C and ARP4754A for flight control — DAL, MC/DC, discretisation verification
7 Flight control	Gain scheduling — design process, frozen-point analysis, LPV methods
8 Engine control	FADEC control architecture — N2 control, surge avoidance, CS-E 650 throttle response
9 All	Control engineering stories — three SAR narratives with aerospace domain translation
10 All / Critical	Portfolio project — NASA GTM flight control law, satellite ADCS, or GPS/IMU filter
11 Flight control	Fly-by-wire architecture — FCC, ADIRU, ACE, latency, triplex voting
12 ADCS roles	Wahba problem and attitude determination — TRIAD, QUEST, star tracker, magnetometer
13 GNC roles	GPS/IMU navigation filter — error-state formulation, 15-state KF, GPS outage
14 Autonomous flight	UAV/eVTOL control architecture — quadrotor PD, trajectory tracking, minimum snap
15 Launch vehicle GNC	Launch vehicle guidance — PEG algorithm, TVC, propellant slosh
16 Flight control	Flight simulation environment — Monte Carlo robustness, Cooper-Harper, HIL testing
17 Flight control / ASE	Aeroelastic flutter — mechanism, V-g diagram, active flutter suppression
18	Quantitative estimation — phase margin, notch filter, LQR gain, Kalman covariance

#	Task Description
All	
19	“Why aerospace — what from control background?” — technically specific 2-minute answer
All	
20	Employer mapping — tool expectations, domain specificity, interview assessment format
All	

THE STRUCTURAL REALITY OF THE CONTROL THEORY-TO-AEROSPACE TRANSITION

The control theory specialist entering aerospace is in the unusual position of having the complete mathematical toolkit for their new domain before they have studied the domain itself. The state space model is the same. The eigenvalue analysis is the same. The Kalman filter derivation is the same. The H-infinity design methodology is the same. What changes is what is on the right-hand side of the equals sign — the physical system being modelled, the physical constraint being respected, the physical failure mode being analysed. The control engineer who has designed an LQR controller for a chemical reactor has designed an LQR controller. The control engineer who then learns what a phugoid mode is and why its time constant constrains the altitude hold bandwidth has designed an LQR controller in a domain that the mathematics has become specific about.

This specificity is the entire preparation task, and it is more achievable than it may appear. The aircraft dynamics that define the plants for flight control are documented in textbooks that are mathematically accessible to any control theory specialist — Etkin, Nelson, Bryson, and Cook all assume mathematical maturity and move quickly to the physical content. The handling quality requirements that define the performance specifications are published documents (MIL-HDBK-1797B is freely available, CS-25 is available from EASA) with numerical values that can be immediately applied to check a design. The spacecraft attitude kinematics that define the ADCS plant are presented in the standard references (Markley and Crassidis is comprehensive and mathematically rigorous) in a way that builds directly on the state space and estimation theory the control engineer already knows.

The portfolio project is the critical differentiator — not because the output is beyond the control engineer's capability, but because completing it transforms the knowledge from understood to applied. The control engineer who has designed a pitch rate command flight control law for the NASA GTM, tuned the short-period response to meet MIL-HDBK-1797B Level 1 requirements, placed the structural mode notch filter to maintain 45° phase margin, and verified the robustness through Monte Carlo simulation across the aerodynamic derivative uncertainty set has done the same engineering work that a first-year flight control engineer does in their first programme assignment. The interview is, at that point, a conversation between two engineers who have worked on the same problem in different contexts — not an assessment of whether the candidate has the mathematics, but a discussion of how they applied it.

The aerospace domain requires the control theory specialist's tools more urgently than any other engineering field. The flight control system must be stable at every point in the flight envelope, robust to aerodynamic uncertainty, tolerant of sensor failures, and certifiable to DO-178C DAL A — simultaneously. The spacecraft attitude controller must function without maintenance for ten years in a radiation and thermal environment that degrades every component. The autonomous flight system must navigate uncertainty in sensor measurements, wind disturbances, and GPS availability while maintaining safe separation from other aircraft. These are genuine, unsolved, practically important control problems — and the control theory specialist who has developed the mathematical tools for robust, optimal, adaptive, and nonlinear control is specifically equipped to work on them. What they need is the aerospace context. This guide provides it.